[DRAFT — for internal circulation]

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RE: A CENTURY AND A HALF OF TREND-FOLLOWING

What a 12-month signal applied to U.S. equities does for return, volatility, and drawdown over 147 years

ABSTRACT ----

We compare a buy-and-hold strategy on the U.S. equity market to a simple trend-following overlay using a 12-month moving average filter, across the full 1871–2018 monthly sample (1,862 monthly observations). The trend strategy invests in equities when the index closes above its 12-month average and in cash otherwise. Over the sample, buy-and-hold returns 9.31% annualised at 14.01% volatility with a maximum drawdown of 81.8% and a Sharpe ratio of 0.66. The trend strategy returns 9.24% annualised at 9.61% volatility with a maximum drawdown of 49.0% and a Sharpe ratio of 0.96. The strategies produce essentially the same long-run return; the trend strategy delivers it with 32% lower volatility, 40% lower maximum drawdown, and a 45% higher risk-adjusted return. The strategy spends 63% of months in equity and 37% in cash. We document the cross-sectional performance across crisis periods, the sub-period robustness of the result, and the cost components (transaction cost, tax drag, opportunity cost during whipsaws) that erode the apparent advantage in real-world implementation.

KEY FINDINGS ----

(1) **Same return, two-thirds of the volatility.** Trend-following matches buy-and-hold's 9.3% annualised return while reducing volatility from 14.0% to 9.6% and Sharpe ratio improves from 0.66 to 0.96.

(2) **Drawdowns are halved.** Maximum drawdown falls from −81.8% (buy-and-hold) to −49.0% (trend). The reduction is concentrated in the 1929–32, 1973–74, and 2007–09 crisis periods.

(3) **63% time in market.** The strategy is invested in equities for 63% of months and in cash for 37%. The cash periods are concentrated around major drawdowns.

(4) **Whipsaws are real but small.** The strategy generates approximately 4–6 round-trip trades per decade. Most are profitable on the entry leg; a meaningful minority produce small losses (whipsaws) of 2–6%.

(5) **Implementation cost matters.** Real-world frictions (transaction cost, bid-ask, tax) reduce the trend advantage by approximately 50–100 basis points per year for taxable accounts. The advantage remains material but is smaller than gross figures suggest.

EXEC. **EXECUTIVE SUMMARY**

> A simple trend-following overlay applied to the U.S. equity market over 147 years achieves what most active strategies promise but few deliver: the same long-run return as buy-and-hold, at substantially lower volatility and dramatically lower drawdown.

This note evaluates a single trend strategy specification: hold equities when the index is above its 12-month average; hold cash (or a cash proxy) otherwise. The signal is observed at month-end and applied to the following month. We test it against a buy-and-hold benchmark across the full Shiller monthly U.S. equity sample, January 1871 through October 2018.

The empirical comparison is striking. Both strategies produce approximately the same annualised return (9.31% buy-hold versus 9.24% trend). The trend strategy achieves this with 32% lower volatility (9.61% vs 14.01% annualised), 40% lower maximum drawdown (49% vs 82%), and a 45% higher Sharpe ratio (0.96 vs 0.66). The result holds in every twenty-year sub-period of the sample.

The mechanism is straightforward. The buy-and-hold strategy participates fully in every drawdown, including those (1929–32, 1973–74, 2000–02, 2007–09) that drove the index down by 50–80% in real terms. The trend strategy exits to cash early in these episodes — typically within 1–3 months of the peak — and re-enters early in the recovery. The exits are imperfect: typically 5–15% below the peak. The re-entries are also imperfect: typically 5–15% above the trough. But the avoided middle portion of each major decline is large enough that the cumulative effect substantially reduces drawdown depth.

The trend strategy has costs that the gross historical numbers understate. Round-trip trading creates transaction costs (modest in modern markets, larger historically). Realised gains create tax obligations in taxable accounts. The strategy generates whipsaws — brief signal flips that produce small losses without preventing larger drawdowns — at a rate of approximately 4–6 per decade. We document these costs explicitly and argue that the post-cost trend advantage is approximately 50–100 basis points per year smaller than the gross advantage we report.

The note is structured as follows. Section 2 specifies the strategy and reviews prior work. Section 3 presents the headline empirical results. Section 4 examines crisis-period performance specifically. Section 5 documents the cost components. Section 6 discusses interpretation, robustness, and implementation considerations.

01. INTRODUCTION

Trend-following has a long pedigree as both a practitioner technique and an academic subject of study. Cowles (1933) tested early forms in U.S. equities and found ambiguous results. Fama (1970) dismissed technical strategies as inconsistent with weak-form market efficiency. Jegadeesh and Titman (1993) documented cross-sectional momentum in equities; Moskowitz, Ooi, and Pedersen (2012) extended the analysis to time-series momentum and found significant Sharpe-improving signals across asset classes.

The contemporary practitioner literature is dominated by managed-futures and CTA strategies that apply trend-following systematically across multiple asset classes (Hamill, Rattray, and Van Hemert, 2016; Bhansali, 2014). The single-asset application — to the U.S. equity index alone — is simpler and arguably more transparent for educational purposes. Faber (2007) documented the strategy in this single-asset specification using a 10-month MA across approximately a century of data; the present note extends and refines that analysis.

This note has three aims. The first is to confirm and extend Faber's original finding through the 2008–2018 decade, including the COVID precursor and post-2008 recovery period. The second is to provide a clean characterisation of the cost components that erode the gross historical advantage in real-world implementation: transaction cost, tax drag, and whipsaw risk. The third is to produce a robust comparison across sub-periods, allowing the reader to assess whether the result is driven by a small number of historical episodes or is structural.

The fundamental claim of the analysis is modest. We do not claim that trend-following produces alpha in the traditional CAPM sense. We claim that it produces equivalent expected return at substantially reduced realised risk, and that the magnitude of the risk reduction is large enough to justify allocation consideration even after implementation costs.

02. METHODOLOGY

2.1 Strategy specification

The strategy is defined as follows. At the close of each month *t*, we observe whether the price index level P_t is above the trailing 12-month moving average MA_t. If P_t > MA_t, the strategy holds equities for month *t+1*, earning the equity total return. If P_t ≤ MA_t, the strategy holds cash for month *t+1*, earning zero (in the base case) or the prevailing short rate (in the appendix variant).

The signal is observable in real-time: the moving average uses only past data, and the position decision for month *t+1* is taken at the close of month *t*. The strategy never looks ahead.

2.2 Benchmarks and metrics

The benchmark is buy-and-hold equities: 100% invested throughout the sample, earning the equity total return monthly. We report:

- **Annualised geometric return.** The standard compound growth measure.
- **Annualised volatility.** Monthly standard deviation × √12.
- **Maximum drawdown.** Worst peak-to-trough decline in the cumulative wealth path.
- **Sharpe ratio.** Annualised return divided by annualised volatility (zero risk-free rate in base case).
- **Time in market.** Fraction of months in which the strategy holds equities.

2.3 Cost modelling

We model implementation cost as: (a) a fixed bid-ask plus commission of 5 basis points per round-trip trade in the modern era (post-1990), graduated upward to 25 bp pre-1990 and 50 bp pre-1950, reflecting empirical historical retail equity transaction costs (Jones, 2002); (b) a tax drag estimate for the taxable case, computed as the long-term capital gains rate (assumed 15%) applied to realised gains at each round-trip; and (c) a whipsaw cost computed implicitly through the strategy returns themselves, since whipsaws produce realised losses captured in the gross numbers.

03. HEADLINE RESULTS

3.1 Full-sample comparison

Table -- **Table 1. Strategy comparison, 1871–2018 (1,862 monthly observations)**

Metric	Buy-and-hold	Trend	Difference
Annualised return	+9.31%	+9.24%	−0.07%
Annualised volatility	14.01%	**9.61%**	−4.40 pp
Sharpe ratio	0.66	**0.96**	+0.30
Maximum drawdown	**−81.8%**	**−49.0%**	+32.8 pp
Time in market	100%	63.3%	—

[Trend Sharpe]	[Buy-hold Sharpe]	[Drawdown reduction]
0.96	**0.66**	**−40%**

The two strategies produce nearly identical long-run returns. The trend strategy achieves this at materially lower risk by every standard measure: 32% lower volatility, 40% lower maximum drawdown, 45% higher Sharpe ratio. The mechanism is essentially redistribution: the strategy concedes some upside (some months in cash that would have been positive) in exchange for avoiding most of the downside (most large drawdown months are correctly identified as below-MA).

3.2 Sub-period stability

Table -- **Table 2. Sub-period performance, 20-year buckets**

Period	B&H ann.	Trend ann.	B&H MDD	Trend MDD
1871–1890	+7.8%	+8.4%	−27%	−13%
1891–1910	+9.0%	+8.8%	−37%	−16%
1911–1930	+10.4%	+10.6%	−47%	−20%
1931–1950	+8.4%	+9.0%	−82%	−32%
1951–1970	+11.8%	+11.5%	−22%	−15%
1971–1990	+10.6%	+10.0%	−47%	−25%
1991–2010	+8.6%	+8.2%	−51%	−21%
2011–2018	+12.7%	+11.9%	−14%	−9%

The pattern is consistent across every 20-year sub-period. Annualised returns are within 1 percentage point in either direction; maximum drawdowns are reduced by roughly half. The 1931–1950 bucket shows the most dramatic drawdown improvement (82% to 32%), driven by the strategy's exit early in the 1929–32 collapse. The 2011–2018 period — a strong bull market — shows the smallest drawdown improvement (the buy-and-hold drawdown was already small), but the trend strategy did not give up substantial returns either.

3.3 Crisis performance

Table -- **Table 3. Crisis-period drawdown comparison**

Crisis	B&H drawdown	Trend drawdown	Avoided
1929–32	−83%	−32%	+51 pp
1937–42	−48%	−18%	+30 pp
1973–74	−47%	−24%	+23 pp
2000–02	−48%	−16%	+32 pp
2007–09	−54%	−21%	+33 pp

The strategy delivers its core value during major drawdowns. In each of the five worst equity crisis periods of the sample, the trend strategy reduced the realised drawdown by 23–51 percentage points relative to buy-and-hold. The 1929–32 case is the most extreme: the buy-and-hold investor lost 83% of their capital; the trend investor lost 32%. The 1937–42 case (less well-known but the second-worst in the sample) similarly saw a 30-percentage-point drawdown reduction.

04. IMPLEMENTATION COSTS

The gross historical numbers reported above understate the cost of implementing the strategy in practice. We treat three cost components: transaction costs, tax drag, and whipsaw frequency.

4.1 Transaction costs

The strategy generates round-trip trades each time the signal flips. The historical signal flip frequency is approximately 4–6 round-trips per decade. Modern (post-1990) round-trip costs for retail S&P 500 ETF execution are approximately 5 basis points (bid-ask plus commission); historical (pre-1990) costs were materially higher, as documented by Jones (2002).

Table -- **Table 4. Estimated transaction cost drag, by era**

Period	RT trades / decade	RT cost (bp)	Annual drag (bp)
1871–1950	5	50	25
1951–1990	4	25	10
1991–2018	5	5	2.5

The annual transaction cost drag has fallen from approximately 25 basis points pre-1950 to under 5 basis points in the modern era. For a contemporary investor, transaction costs are not a meaningful obstacle to trend-following implementation.

4.2 Tax drag

For taxable accounts, realised gains are subject to capital gains tax. The trend strategy realises gains each time it exits to cash (assuming the entry price was lower than the exit price), creating a current tax obligation that buy-and-hold defers indefinitely. Assuming a 15% long-term capital gains rate and an average gain of 8% per round-trip, the tax drag is approximately 60 basis points per round-trip, or roughly 30 basis points per year given typical signal-flip frequencies.

For tax-advantaged accounts (IRA, 401(k), or international equivalents), this drag does not apply. The strategy is therefore materially more attractive for sheltered accounts than for taxable ones.

4.3 Whipsaws

A whipsaw is a signal pattern in which the strategy exits, re-enters, and exits again in close succession, producing small losses without preventing a larger drawdown. We define a whipsaw as a sequence in which the strategy exits and re-enters within three months without the price reaching a new low between the exit and re-entry.

The historical whipsaw frequency is approximately 1.5 whipsaws per decade in our sample. Each whipsaw produces an average loss of approximately 3% (relative to having stayed in equities). Total whipsaw cost is therefore approximately 0.45% per year. This is implicitly captured in our gross strategy returns, but is worth noting separately because investors experiencing whipsaws often abandon the strategy and lock in the losses without capturing the subsequent (typically larger) avoided drawdowns.

4.4 Net advantage

Table -- **Table 5. Estimated net trend advantage by account type**

Account type	Gross trend Sharpe gap	Net (after costs)
Tax-advantaged, modern era	+0.30	+0.27
Taxable, modern era	+0.30	+0.20
Tax-advantaged, mid-century era	+0.30	+0.22
Taxable, mid-century era	+0.30	+0.15

The strategy's risk-adjusted advantage net of all implementation costs ranges from +0.15 to +0.27 in Sharpe ratio terms. This is meaningful — equivalent to a 20–40% improvement in risk-adjusted returns relative to buy-and-hold — but is materially smaller than the gross +0.30 advantage. Practitioners should not be surprised to see live performance slightly weaker than backtested gross performance.

05. DISCUSSION, LIMITATIONS, AND CONCLUSION

What the strategy does and does not deliver. The strategy does not produce alpha — it does not outperform buy-and-hold on absolute return. It produces what financial economics would call risk-managed beta: the same beta-equivalent return at lower risk. For an investor whose objective is risk-adjusted long-run wealth, this is a meaningful improvement. For an investor whose objective is maximum absolute returns at any cost, it is not.

Behavioural fit. The strategy spends 37% of months in cash. In long stretches of bull-market activity, this can produce extended periods (24–60 months) in which trend underperforms buy-and-hold materially. The 2017–2018 period is a recent example: buy-and-hold returned approximately 25% over the two years; trend returned approximately 22% (the small underperformance reflects two brief whipsaws). Investors prone to abandon strategies during such periods will not capture the long-run benefit, which is concentrated in crisis periods that occur infrequently.

Limitations. Our analysis is U.S.-only; international results are qualitatively similar but with somewhat smaller magnitudes (Faber, 2013). We use a single signal (12-month MA); ensembles of signals (12-month plus 6-month, momentum plus volatility) tend to produce smoother results but qualitatively similar magnitudes. We assume zero return on cash in the base case; appendix A2 reports results with a contemporaneous short-rate assumption. Most importantly, our analysis does not address single-stock trend-following, which has different (and generally less attractive) characteristics.

Conclusion. A simple 12-month trend-following overlay on the U.S. equity market produces equivalent long-run returns to buy-and-hold at substantially lower volatility and dramatically lower maximum drawdown. The result is robust across 147 years of monthly data, holds in every twenty-year sub-period, and is consistent with a substantial body of academic literature. Implementation costs erode the gross advantage modestly but do not eliminate it. The strategy is a reasonable component of a diversified investment approach, particularly in tax-advantaged accounts, but is not a substitute for sound asset allocation.

>> Trend-following does not generate alpha. It generates the ***same*** return at ***lower*** risk. For an investor whose objective is compound wealth subject to behavioural constraints, that is the more useful improvement.

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APX. APPENDIX A: ROBUSTNESS CHECKS

A.1 Window length sensitivity

Table -- **Table A1. Strategy performance across moving-average windows**

Window	Ann. return	Ann. vol	Max DD	Sharpe
6-month	+8.91%	10.31%	−42.1%	0.86
10-month	+9.32%	9.83%	−47.6%	0.95
12-month (base)	+9.24%	9.61%	−49.0%	0.96
14-month	+9.18%	9.78%	−51.3%	0.94
24-month	+8.74%	10.74%	−54.9%	0.81

The 10–14 month range produces near-optimal Sharpe ratios. Shorter windows (6-month) generate more whipsaws, modestly degrading return and Sharpe. Longer windows (24-month) react more slowly to drawdowns, allowing larger maximum losses.

A.2 With short-rate cash return

The base-case analysis assumes zero return on cash. Allowing cash to earn the contemporaneous Shiller short rate yields:

Table -- **Table A2. With short-rate cash return**

Specification	Ann. return	Sharpe
Trend, zero cash	+9.24%	0.96
Trend, with short rate	+10.61%	1.10

Allowing cash to earn the prevailing short rate boosts trend's annualised return by approximately 140 basis points and improves the Sharpe ratio further. Net of historical transaction costs, the improvement is approximately 100 bp.

A.3 Combination strategies

Table -- **Table A3. Trend ensemble strategies**

Strategy	Ann. return	Max DD	Sharpe
12-month MA only	+9.24%	−49.0%	0.96
10/12-month MA average	+9.21%	−47.5%	0.97
6/12-month MA average	+9.08%		0.95

A.4 ProRealTime: 12-month trend-following signal

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Time-series momentum, 12-month lookback
// Daily bars on multi-asset basket; one instance per instrument

DEFPARAM CumulateOrders = False
DEFPARAM PreLoadBars = 270

lookback = 252  // ~12 months of trading days
ref_price = close[lookback]

// Long if 12-month return positive, flat otherwise
mom = (close - ref_price) / ref_price

IF mom > 0 AND NOT LongOnMarket THEN
  BUY 1 CONTRACT AT MARKET
ENDIF
```

Strategy	Ann. return		Max DD	Sharpe

		IF mom <= 0 AND LongOnMarket THEN
		  SELL AT MARKET
		ENDIF
		
		// For the volatility-targeted variant, scale position size by 1/recent ATR
		// See companion paper "Position sizing dominates strategy selection" (Mar 2020)
		="num">−46.2%		

3-signal ensemble	+9.18%		−45.8%	0.97

Ensemble strategies produce marginally higher Sharpe ratios at the cost of additional implementation complexity. For most practical purposes, the single 12-month signal captures most of the available risk-adjusted improvement.

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